

Tunneling

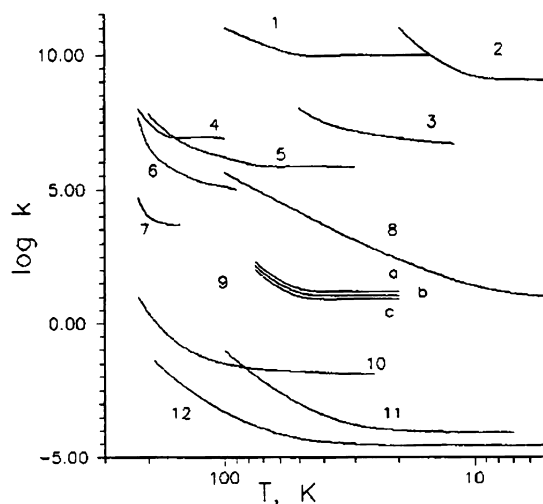
“...if the de Broglie wave gets through, even though with some difficulty, it always smuggles a particle with it. ... To evaluate this formula I had to calculate the integral of the expression $\sqrt{1-a/r}dr$, and I did not know how to do it. ... I expressed at the end of [the paper] my thanks to Kotshchin for his help with the mathematics. Later when the paper appeared, he wrote me that he had become a laughingstock among his colleagues, who had learned what kind of highbrow mathematical help he had given me.”

Gamow, My World Line (1970)

Tunneling [1] refers to transmission through a barrier at energies below the classical reaction energy threshold. The hallmark of tunneling is a rate that becomes temperature independent at low temperatures. Tunneling can usually (but not always [2]) be ignored at high temperatures or when heavy atoms are important in the reaction coordinate. But for reactions involving proton or hydrogen transfer, tunneling can dramatically accelerate rates even at room temperature. Figure 12.0.1 from Benderskii et al. [3] compiles data for many reactions that exhibit strong tunneling contributions, some at temperatures as high as 200K. The figure also shows that the transition between thermally activated and tunneling dominated regimes can take place over a wide temperature interval.

The onset of tunneling can also be seen on an Arrhenius plot as the temperature interval where $\ln k$ vs. $1/k_B T$ changes from a slope of $-E_a$ to a slope of approximately zero. When only a small temperature interval is examined, the effects of tunneling are easily misinterpreted as artificially small activation energy or enthalpy parameters. At temperatures where tunneling is important, the experimental activation parameters are inextricably mixed with tunneling contributions. There is no way to change this situation, because the experiment cannot “turn off” tunneling to obtain a classical barrier estimate. Ignoring tunneling can lead to errors in certain applications, e.g. in testing new mechanisms by comparing predicted and observed activation enthalpies, or for identifying the rate determining step in a multistep reaction with proton or hydrogen transfer steps. For reactions that involve H -atom motion at moderate or low temperatures, the best practice is to check for the possible importance of tunneling. This chapter discusses simple tests and corrections to account for tunneling.

Figure 12.0.1: Tunneling leads to temperature independent rates below a tunneling crossover temperature. Signatures of tunneling have been observed for many reactions as shown in this compilation of results for several low temperature reactions. [From Benderskii et al. *Physics Reports*, 233, 195–339 (1993).]



12.1 One-dimensional tunneling models

Real molecules are never one-dimensional, but one-dimensional tunneling models are often surprisingly effective. One dimensional tunneling models assume the reaction coordinate is separable from all other degrees of freedom. In the vicinity of the saddle point, recall that the unstable mode coordinate of harmonic TST has locally separable dynamics. The one dimensional classical TST rate constant for passage over a barrier of height $V_{\ddagger}$ is $k_{TST} = k_B T h^{-1} \exp[-V_{\ddagger}/k_B T]$. For one isolated degree of freedom, the TST rate constant can equally be written as [4]

$$k_{TST} = \frac{1}{h} \int_0^{\infty} P_{CL}(E) e^{-E/k_B T} dE \quad (12.1.1)$$

$$= \frac{k_B T}{h} \int_0^{\infty} P_{CL}(E) e^{-E/k_B T} d(E/k_B T) \quad (12.1.2)$$

where the lower integration bound starts at the zero point energy level in the reactant state and where the factor P_{CL} is the classical transmission probability

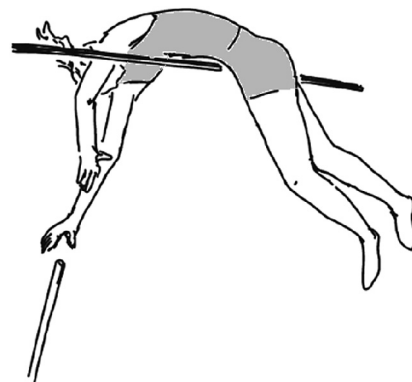
$$P_{CL}(E) = \begin{cases} 1 & \text{for } E > V_{\ddagger} \\ 0 & \text{for } E < V_{\ddagger} \end{cases} \quad (12.1.3)$$

The physics behind equation (12.1.3) is trivial. A classical point particle moving along the q -axis is reflected by the barrier if the energy is $E < V_{\ddagger}$ and transmitted if the energy is $E > V_{\ddagger}$.

Quantum mechanically, the same particle would be delocalized over some interval on the q -axis. Even in the classical world, a delocalized body can do something akin to tunneling

over a barrier. For example, consider the pole-vaulter in Figure 12.1.1. Her center of mass never eclipses the barrier threshold, and yet she clears the bar anyway. Of course, the pole vaulter differs from quantum mechanical tunneling in many aspects, but some useful models of tunneling (especially ring-polymer molecular dynamics) exhibit barrier crossing trajectories much like those of the pole-vaulter.

Figure 12.1.1: A good pole vaulter can outjump her own center-of-mass by contorting her body as it goes over the bar. A ring polymer tunnels over reaction barriers in the same manner.



Quantum mechanically, a trajectory can tunnel through the barrier at energies below $V_{\ddagger}$ and trajectories can also be reflected at energies above $V_{\ddagger}$. Given a model for the tunneling probability $P_{QM}(E)$, the rate constant is [4,5]

$$k = \frac{k_B T}{h} \int_0^{\infty} P_{QM}(E) e^{-E/k_B T} d(E/k_B T). \quad (12.1.4)$$

It is usually convenient to define a transmission coefficient κ_{QM} that isolates the effects of tunneling. The transmission coefficient is the ratio of the rates with (equation (12.1.4)) and without (equation (12.1.1)) the effects of tunneling.

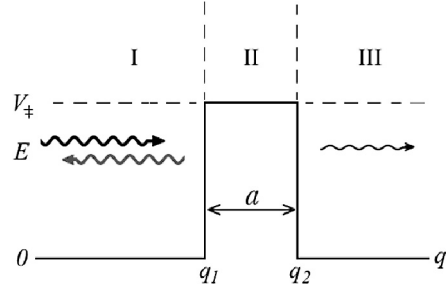
$$\begin{aligned} \kappa_{QM} &= \frac{\int_0^{\infty} P_{QM}(E) e^{-E/k_B T} d(E/k_B T)}{\int_0^{\infty} P_{CL}(E) e^{-E/k_B T} d(E/k_B T)} \\ &= \int_0^{\infty} P_{QM}(E) e^{-(E-V_{\ddagger})/k_B T} d(E/k_B T) \end{aligned} \quad (12.1.5)$$

Thus far, this book has avoided serious quantum mechanics, but at least a minimal introduction is needed to explain the nature of the tunneling probability $P_{QM}(E)$ and to obtain quantitative expressions for κ_{QM} .

A standard instructive calculation [4] considers tunneling through a square-top barrier of height $V_{\ddagger}$. It is customary since the work of Gamow [6], but somewhat counterintuitive, to derive the transmission and reflectance probabilities (see Figure 12.1.2) using a time independent

Schrodinger equation.¹ How can a time-independent equation describe the incoming trajectories and the subsequent transmission and reflection amplitudes? Welcome to the odd world of quantum mechanics! The short answer is that a continuous beam of scatterers is no different from an individual (as long as they have the same energy and cannot interfere with each other). A better answer requires wavepacket scattering calculations [8], and those are beyond the scope of this book.

Figure 12.1.2: Tunneling through a square-top barrier of height $V_{\ddagger}$.



The quantum mechanical Hamiltonian is

$$\hat{H} = -\frac{\hbar}{2m} \frac{\partial^2}{\partial q^2} + V(q) \quad (12.1.6)$$

and the time-independent Schrodinger equation is

$$\hat{H}\Psi = E\Psi. \quad (12.1.7)$$

In regions I and III, the potential is $V(q) = 0$, and the momentum is a constant given by $E = p^2/2m$. The wavefunctions, with as yet undetermined coefficients, are a superposition of left (−) and right (+) moving waves:

$$\Psi_I(q) = A_{I+}e^{ipq/\hbar} + A_{I-}e^{-ipq/\hbar} \quad (12.1.8)$$

$$\Psi_{III}(q) = A_{III+}e^{ipq/\hbar} + A_{III-}e^{-ipq/\hbar} \quad (12.1.9)$$

In region II, the potential is $V(q) = V_{\ddagger}$, and for $E < V_{\ddagger}$ the wavefunction is

$$\Psi_{II}(q) = A_{II+}e^{|p_{CL}|q/\hbar} + A_{II-}e^{-|p_{CL}|q/\hbar} \quad (12.1.10)$$

where the classical momentum is imaginary

$$p_{CL} = \sqrt{2m(E - V_{\ddagger})} \quad (12.1.11)$$

¹ See Merzbacher's article on the history of tunneling [7].

Equations (12.1.8), (12.1.9), and (12.1.10) are easily verified solutions to equation (12.1.7) in the respective regions. Continuity and differentiability are requirements of an admissible wavefunction for a bounded potential. Thus we must choose the six coefficients so that Ψ is continuous and differentiable throughout. The matching conditions are thus $\Psi_I(q_1) = \Psi_{II}(q_1)$, $\Psi'_I(q_1) = \Psi'_{II}(q_1)$, $\Psi_{II}(q_2) = \Psi_{III}(q_2)$, and $\Psi'_{II}(q_2) = \Psi'_{III}(q_2)$. Four matching equations cannot identify six unknown coefficients, so two additional conditions are needed. One more constant can be specified based on the nature of the incident wave. If the incident trajectory is moving toward the right, it cannot result in a leftward traveling wave in region III, so $A_{III-} = 0$. Additionally, the amplitude of the incident wave is arbitrary, i.e. we are only concerned with the *fractional* transmittance and reflectance. For convenience, take the incident amplitude as $A_{I+} = 1$. Now there are six equations and six unknowns. After some algebra, the reflected and transmitted wave amplitudes can be obtained. The calculation can be repeated for cases where $E > V_{\ddagger}$ in which case the wave function is oscillatory in sections I, II, and III. After some more algebra to match solutions, the transmittance and reflectance amplitudes emerge also for classically allowed energies. In each case, $|A_{III+}|^2$ gives the transmittance probability as a function of E :

$$P_{QM}(E) = \begin{cases} \left[1 + \frac{V_{\ddagger}^2}{4E(E-V_{\ddagger})} \sin^2\left(\frac{a}{\hbar} \sqrt{2m(E-V_{\ddagger})}\right) \right]^{-1} & E > V_{\ddagger} \\ \left[1 + \frac{V_{\ddagger}^2}{4E(V_{\ddagger}-E)} \sinh^2\left(\frac{a}{\hbar} \sqrt{2m(V_{\ddagger}-E)}\right) \right]^{-1} & E < V_{\ddagger} \end{cases} \quad (12.1.12)$$

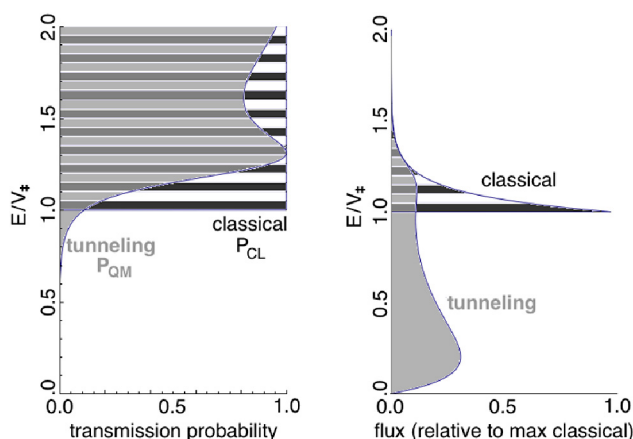
with the probability of reflection being $1 - P_{QM}(E)$. In the classically forbidden case ($E < V_{\ddagger}$), the quantity $2a\sqrt{2m(V_{\ddagger}-E)}$ that appears within $\sinh$ is the classical action for one orbit in an inverted (upside-down) version of the barrier in Figure 12.1.2.

Contributions to the rate constant at each energy are proportional to $P_{QM}(E)$ and also to the Boltzmann factor $\exp[-E/k_B T]$ as per equation (12.1.4). Figure 12.1.3 shows the contributions to k including tunneling and also the classical contributions to k_{TST} . It is convenient to write the tunneling probability in terms of dimensionless groups and familiar properties. The parameters that govern tunneling are the incoming energy relative to the barrier height and the width of the barrier relative to the thermal de Broglie wavelength. The barrier height in units of $k_B T$ is also important for the Boltzmann factor. Therefore it is useful to write equation (12.1.12) in terms of $\lambda_T = \hbar/\sqrt{2\pi m k_B T}$, $u_0 = V_{\ddagger}/k_B T$, and $\varepsilon = (E - V_{\ddagger})/V_{\ddagger}$.

$$P_{QM}(\varepsilon) = \begin{cases} \left[1 + (4(\varepsilon + 1)\varepsilon)^{-1} \sin^2\left(\frac{a}{\lambda_T} \sqrt{u_0 \varepsilon / \pi}\right) \right]^{-1} & \varepsilon > 0 \\ \left[1 - (4(\varepsilon + 1)\varepsilon)^{-1} \sinh^2\left(\frac{a}{\lambda_T} \sqrt{u_0 \varepsilon / \pi}\right) \right]^{-1} & \varepsilon < 0 \end{cases} \quad (12.1.13)$$

The square top barrier is a good model for certain applications, e.g. electron tunneling through a molecular spacer or insulating layer. It is a rather poor model for proton transfer reactions, but we can use it to illustrate qualitative aspects of the tunneling probability. Recall from the discussion of TST that the thermal de Broglie wavelength at 300K and for an effective mass of 1amu is approximately $1\text{\AA}$. A representative distance a for proton transfer between donor and acceptor in a chemical reaction is around $3\text{\AA}$. Therefore, the values $u_0 = 12$ and $(a/\lambda_T)\sqrt{u_0/\pi} = 10$ are representative. Figure 12.1.3 below shows the classical and quantum transmission probability and also the integrands $\exp[-E/k_B T]P_{CL}(E)$ and $\exp[-E/k_B T]P_{QM}(E)$. There is some propensity for non-classical reflection. However, multiplication of $P_{QM}(E)$ by $\exp[-E/k_B T]$ in equation (12.1.4) accentuates low energy phenomena making non-classical tunneling far more important than non-classical reflection.

Figure 12.1.3: The transmission probability $P_{QM}(E)$ and the Boltzmann weighted contribution to the flux through the square-top barrier, $\exp[-E/k_B T]P_{QM}(E)$. Representative dimensionless parameter values $u_0 = 12$ and $(a/\lambda_T)\sqrt{u_0/\pi} = 10$ were used to construct the plots.



The square-top barrier has uniform width from top to base, and the narrow base makes it especially permeable to deep tunneling at very low energies. The propensity for deep tunneling can be seen in Figure 12.1.3. Barriers for real chemical reactions tend to broaden at the base, so that the onset of tunneling is more gradual as temperature is lowered.

The calculation above is a quantum scattering calculation for a square topped one-dimensional barrier. The quantum mechanical scattering problem for real molecular potentials is far more difficult and worthy of an entirely separate book [9]. Fortunately, the Wentzel-Kramers-Brillouin (WKB) approximation [10] gives adequate estimates of the tunneling probability for hydrogen nuclei. The tunneling probability [6] according to a WKB approximation is

$$P_{SC} = \{1 + \exp[2\theta(E)]\}^{-1} \quad (12.1.14)$$

where

$$\theta(E) = \frac{1}{\hbar} \int_{q_1(E)}^{q_2(E)} \sqrt{2m|V(q) - E|} dq \quad (12.1.15)$$

is a dimensionless barrier penetration integral. The subscript “SC” refers to semiclassical mechanics of which the WKB approximation is a special one-dimensional case. $q_1(E)$ and $q_2(E)$ are classical turning points which define the left and right edges of the classically forbidden region. The barrier penetration integral is the classical action for an orbit between $q_1(E)$ and $q_2(E)$ when the potential is turned upside down [11,12]. Recall that a similar classical-action-on-inverted-potential quantity emerged in the quantum transmission probabilities for the square top barrier [11,12]. To obtain the semi-classical transmission coefficient, use the WKB equation (12.1.5) with equation (12.1.14)

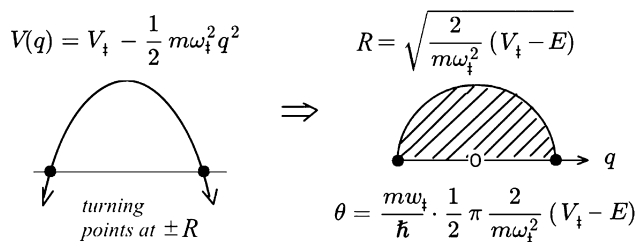
$$\kappa_{QM} = \int_0^\infty P_{SC}(E) e^{-(E-V_\ddagger)/k_B T} d(E/k_B T) \quad (12.1.16)$$

where $E = 0$ corresponds to the reactants in their ground state energy. Equation (12.1.16) is a semiclassical approximation to the true quantum mechanical results. Some results for realistic barrier models are outlined below.

Parabolic barrier

In the same year (1932) of his seminal TST calculation [13], Wigner developed an approximation to the tunneling correction for a parabolic barrier [14]. The Wigner model approximates the barrier as an infinite downward parabola having an imaginary frequency with magnitude $\omega_\ddagger$. If the parabolic barrier is sufficiently broad, i.e. if $\hbar|\omega_\ddagger|/k_B T \ll 1$ is sufficiently small, then all tunneling will occur near the barrier top making details of the ground state irrelevant. To evaluate κ_{QM} we must first determine the barrier penetration integral $\theta(E)$. For the parabolic barrier, θ is proportional to the area under a half-circle as shown in Figure 12.1.4.

Figure 12.1.4: For a parabolic barrier, $\theta(E)$ corresponds, within constants, to half of the area under a circle with radius $R = (2(V_\ddagger - E)/(m\omega_\ddagger^2))^{(1/2)}$.



Following the calculation depicted in Figure 12.1.4, the barrier penetration integral as a function of energy for the parabolic barrier is

$$\theta(E) = \frac{\pi(V_\ddagger - E)}{\hbar\omega_\ddagger} \quad (12.1.17)$$

Using equation (12.1.17) in equation (12.1.16), we can change the variable of integration to θ . The change of variables gives

$$\kappa_{QM} = \int_{-\infty}^{\infty} \frac{e^{\alpha_0 \theta / \pi}}{1 + e^{2\theta}} \frac{\alpha_0}{\pi} d\theta \quad (12.1.18)$$

where $\alpha_0 = \beta \hbar \omega_{\ddagger}$. The mass appears to have vanished from the calculation, but it is still there within $\omega_{\ddagger}$ because different isotopes will give different imaginary frequencies. More will be said about this point in the discussion of isotope effects. At temperatures where $\alpha_0 < 2\pi$, the integral can be done exactly, giving

$$\kappa_{QM}(T) = \frac{\alpha_0/2}{\sin(\alpha_0/2)} \quad (12.1.19)$$

Equation (12.1.19), the Bell tunneling correction [15], is the closed form result from an infinite parabolic barrier model.

For low temperatures where $\alpha_0 > 2\pi$, the integrand in equation (12.1.18) diverges as $\theta \rightarrow \infty$. This is an artifact from extending the parabolic potential to $E \rightarrow -\infty$. Wigner had previously suggested the approximation

$$\kappa_{QM}(T) \approx 1 + \alpha_0^2/24 \quad (12.1.20)$$

The Wigner correction is just the first two terms in a series expansion for the exact formula: $\kappa_{QM}(T) = 1 + \alpha_0^2/24 + 7\alpha_0^4/5760 + \dots$. Wigner's correction can be computed for all temperatures, but the extended temperature range is achieved by truncating a series that diverges. The table below shows the Bell and Wigner formulas begin to diverge from each other for $\alpha_0 > \pi$, a sign that the higher order terms are already becoming important. The rightmost column compares the third and second terms of the expansion in $\beta \hbar \omega_{\ddagger}$.

α_0	$\frac{\alpha_0/2}{\sin(\alpha_0/2)}$	$1 + \frac{\alpha_0^2}{24}$	$\frac{7\alpha_0^4/5760}{\alpha_0^2/24}$	
0.1	1.00	1.00	0.0003	Bell more justified than Wigner
1.0	1.04	1.04	0.03	
3.2	1.58	1.42	0.3	
5.0	4.22	2.04	0.7	do not use Bell or Wigner
6.3	nonsense	2.66	1.2	

Neither the Wigner nor Bell versions of the parabolic barrier correction can qualitatively capture the low temperature rate behavior shown in Figure 12.0.1. Johnston and Rapp [16] recommend that the parabolic barrier tunneling corrections should only be used when $\beta \hbar \omega_{\ddagger} < 3.6$. Unfortunately, the temperatures at which the simple Bell and Wigner corrections are valid coincide with temperatures at which tunneling is largely unimportant. In practice, these are mostly useful as

tests to gauge the importance of tunneling. If $\beta\hbar\omega_{\ddagger} > \pi$ then tunneling is important, and one of the more accurate tunneling models below should be used for accurate rates and activation parameters.

Shortcomings of the parabolic barrier model motivated several alternative one-dimensional tunneling models for low temperatures. The alternative calculations vary widely in the amount of information about the potential energy surface that is required. The most accurate corrections require the complete minimum energy pathway (MEP) and the vibrational frequencies of the transverse modes along the MEP [17]. Examples include direct applications of WKB theory using the potential as a function of arclength along the MEP [18] and the vibrationally adiabatic tunneling model [19]. The simplest low-temperature one-dimensional models of tunneling require no more information than what is already required for a harmonic TST calculation. These include the truncated parabolic barrier model [20,21] and corrections based on an Eckart barrier model [22–24].

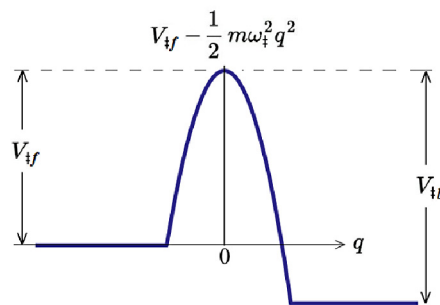
The truncated parabolic barrier

The parabolic barrier tunneling correction diverged at low temperatures because the ground state energy was artificially extended to $E \rightarrow -\infty$. The truncated parabolic barrier model fixes the unphysical deep tunneling artifacts by replacing the lower integration bound in equation (12.1.18) with the ground state energy [20]. The truncated parabolic barrier is an extremely simple tunneling correction that remains well defined (but not necessarily accurate) at arbitrarily low temperatures. It can be derived using the same penetration integral (equation (12.1.17)) from the analysis of the infinite parabolic barrier.

Suppose that the force constant along the reaction coordinate at the barrier top is $\partial^2 V / \partial q^2|_{\ddagger} = -m\omega_{\ddagger}^2$. Also suppose that the forward and backward activation energy thresholds have been determined: $V_{\ddagger f}$ and $V_{\ddagger b}$. These should already account for zero point energies of the reactants, products, and transition states. The truncated parabolic barrier model based on parameters $V_{\ddagger f}$, $V_{\ddagger b}$, and $m\omega_{\ddagger}^2$ is shown in Figure 12.1.5. Note that tunneling changes rates, but it should not change equilibrium constants. The TST rate constant already satisfies detailed balance, so tunneling corrections for the forward and backward directions must be equivalent. A thermodynamically consistent tunneling correction can be obtained by computing one tunneling correction from the smaller of the forward and backward barriers. Here we assume that $V_{\ddagger f} < V_{\ddagger b}$.

The parabolic barrier is constructed to match the zero-point corrected activation energy in the forward and reverse directions as well as the second derivative of the potential along the reaction coordinate at the transition state. Where entrance or exit complexes have lower energy than

Figure 12.1.5: The truncated parabolic barrier model eliminates spurious tunneling contributions from energies below the reactant and product ground states. The tunneling correction is computed for the exothermic direction (the forward direction in the figure).



infinitely separated reactants or products, the energies of the entrance and exit complexes should be used to define $V_{\ddagger f}$ and $V_{\ddagger b}$ [25].

The truncated parabolic barrier tunneling correction is a function of two dimensionless parameters $\alpha_0 = \beta \hbar \omega_{\ddagger}$ and $\alpha_1 = 2\pi V_{\ddagger f} / \hbar \omega_{\ddagger}$:

$$\kappa_{QM} = \frac{\alpha_0/2}{\sin[\alpha_0/2]} - \frac{\alpha_0/2\pi}{1 - \alpha_0/2\pi} \exp\left[\alpha_1 \left(\frac{\alpha_0}{2\pi} - 1\right)\right] \quad \text{for } \alpha_0 \leq 2\pi \quad (12.1.21)$$

or

$$\kappa_{QM} = \frac{\alpha_0/2\pi}{\alpha_0/2\pi - 1} \left\{ \exp\left[\alpha_1 \left(\frac{\alpha_0}{2\pi} - 1\right)\right] - 1 \right\} \quad \text{for } \alpha_0 \geq 2\pi \quad (12.1.22)$$

The piecewise transmission coefficient formula is a continuous smooth function of α_0 . The truncated parabolic barrier correction tends to overestimate the importance of tunneling because the parabolic barrier is artificially narrow at low energies [26]. Nevertheless, it correctly describes the vanishing activation energy at low temperatures.

The asymmetric Eckart barrier

Eckart introduced a simple one dimensional barrier model for which the tunneling and reflectance probabilities could be determined exactly in the symmetric case [22]. Johnston and Heicklen later obtained tunneling corrections for the asymmetric Eckart model of exothermic and endothermic reactions [23]. Eckart models are usually the most accurate among those that do not require information about the potential energy surface beyond what is required by harmonic TST. The Eckart barrier has the form [22]

$$V(q) = \frac{Ay}{1+y} + \frac{By}{(1+y)^2} \quad (12.1.23)$$

where

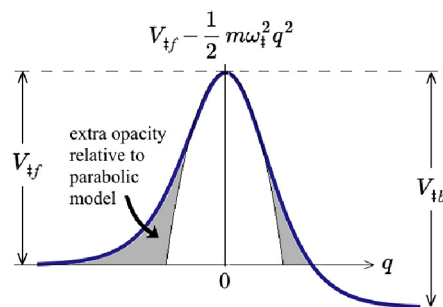
$$y = \exp[2\pi q/L] \quad (12.1.24)$$

A , B , and L are defined in terms of $V_f^\ddagger$, $V_b^\ddagger$, and the force constant along the reaction coordinate at the barrier top, $\partial^2 V / \partial q^2|_\ddagger = -m\omega_\ddagger^2$.

$$\begin{aligned} A &= V_{\ddagger f} - V_{\ddagger b} \\ B &= \left\{ \sqrt{V_{\ddagger f}} + \sqrt{V_{\ddagger b}} \right\}^2 \\ \frac{L}{2\pi} &= \left\{ \sqrt{m\omega_\ddagger^2/2V_{\ddagger f}} + \sqrt{m\omega_\ddagger^2/2V_{\ddagger b}} \right\}^{-1} \end{aligned} \quad (12.1.25)$$

Again, the barriers $V_{\ddagger f}$ and $V_{\ddagger b}$ should already account for zero point energies of the reactants, products, and transition states. And again, where entrance or exit complexes have lower energy than infinitely separated reactants or products, the energies of the entrance and exit complexes should be used to define $V_{\ddagger f}$ and $V_{\ddagger b}$ [25]. Figure 12.1.6 shows the Eckart barrier model along with a (truncated) parabolic barrier model for the same reaction.

Figure 12.1.6: The asymmetric Eckart potential gives a realistic interpolated barrier that matches the zero-point corrected activation energy in the forward and reverse directions as well as the second derivative of the potential along the reaction coordinate at the transition state.



The relationships between $V_{\ddagger f}$, $V_{\ddagger b}$, and $m\omega_\ddagger^2$ and the Eckart potential parameters A , B , and L can also be inverted

$$\begin{aligned} m\omega_\ddagger^2 &= \pi^2(A^2 - B^2)/(2L^2B^2) \\ V_{\ddagger f} &= (A + B)^2/4B \\ V_{\ddagger b} &= (A - B)^2/4B \end{aligned} \quad (12.1.26)$$

Equations (12.1.25) are useful for converting *ab initio* results to an interpolating Eckart potential. Equations (12.1.26) are useful for comparing the Eckart barrier tunneling correction to results from other models.

As for the truncated parabolic barrier correction, the asymmetric Eckart barrier tunneling correction is based on the smaller of the forward and backward activation barriers. The same tunneling correction then applies to both the forward and backward TST rate constants. These conventions ensure detailed balance for the tunneling corrected rate constants. In the usual harmonic TST approximation starting from mass scaled coordinates, the appropriate mass is

simply $m_0 = 1 \text{ amu}$. If other coordinates are used, procedures outlined by Garrett and Truhlar [27] or Johannesson and Jonsson [28] should be used to find the appropriate mass. As Garrett and Truhlar have emphasized, the results do not depend on conventions for scaling the reaction coordinate as long as the mass is correspondingly scaled [27].

The probability that a trajectory will tunnel through an Eckart barrier as a function of the incident energy E is

$$P_{QM}(E) = 1 - \frac{\cosh[2\pi(a-b)] + \cosh[2\pi d]}{\cosh[2\pi(a+b)] + \cosh[2\pi d]} \quad \text{if } d \in \mathbb{R} \quad (12.1.27)$$

or

$$P_{QM}(E) = 1 - \frac{\cosh[2\pi(a-b)] + \cos[2\pi|d|]}{\cosh[2\pi(a+b)] + \cos[2\pi|d|]} \quad \text{if } d \notin \mathbb{R} \quad (12.1.28)$$

where

$$2\pi a = 2(\alpha_1 \xi)^{1/2} (\alpha_1^{-1/2} + \alpha_2^{-1/2})^{-1} \quad (12.1.29)$$

$$2\pi b = 2\{\alpha_1 \xi + |\alpha_1 - \alpha_2|\}^{1/2} (\alpha_1^{-1/2} + \alpha_2^{-1/2})^{-1} \quad (12.1.30)$$

$$2\pi d = 2(\alpha_1 \alpha_2 - \pi^2/4)^{1/2} \quad (12.1.31)$$

and where

$$\alpha_1 = 2\pi V_{\ddagger f} / \hbar \omega_{\ddagger} \quad (12.1.32)$$

$$\alpha_2 = 2\pi V_{\ddagger b} / \hbar \omega_{\ddagger} \quad (12.1.33)$$

$$\xi = E / V_{\ddagger f} \quad (12.1.34)$$

Don't be alarmed by all of those definitions. They are all emerging from the incident energy E and just three barrier characteristics $V_{\ddagger f}$, $V_{\ddagger b}$, and $m\omega_{\ddagger}^2$. It is convenient to think of $P_{QM}(E)$ as a function $P_{QM}(\xi | \alpha_1, \alpha_2)$ in which α_1, α_2 are parameters of the potential, and ξ will be integrated out in the calculation of κ_{QM} . Note that Johnston and Heicklen contains a critical typo in the equation for parameter b [23]. The formulas above work correctly provided the convention $V_{\ddagger f} < V_{\ddagger b}$ is maintained.

While the tunneling probability P_{QM} is exact (for the model potential), the tunneling transmission coefficient κ_{QM} must be computed numerically. This is not a major disadvantage. Numerical integration is required for all but the simplest parabolic barrier models. The tunneling transmission coefficient, which also depends on the usual parameter $\alpha_0 = \beta \hbar \omega_{\ddagger}$, is

$$\kappa_{QM} = \frac{\alpha_1 \alpha_0}{2\pi} \int_0^\infty \exp \left[-\frac{(\xi - 1) \alpha_1 \alpha_0}{2\pi} \right] P_{QM}(\xi | \alpha_1, \alpha_2) d\xi \quad (12.1.35)$$

The integration is fairly straightforward, but Figure 12.1.7 provides a table of computed values in terms of the three dimensionless parameters α_1 , α_2 , and $\alpha_0 = \beta\hbar\omega_{\ddagger}$. An early application of the Eckart tunneling correction can be seen in the work by Shultz and LeRoy [29] on $H + H_2 \rightleftharpoons H_2 + H$, one of the first studies to clearly show a curved Arrhenius plot.

Figure 12.1.7: Tabulated values of $\ln(\kappa_{QM})$ from the asymmetric Eckart barrier as a function of α_1 , α_2 , and $\beta\hbar\omega_{\ddagger}$. In each graph, moving to the right corresponds to increasing $V_{\ddagger b}$, and moving downward corresponds to increasing $V_{\ddagger f}$. The diagonal corresponds to results for a symmetric Eckart barrier.

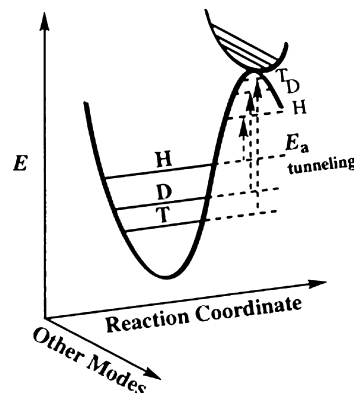
$\beta\hbar\omega_{\ddagger} = 2$					$\beta\hbar\omega_{\ddagger} = 10$				
$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0	$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0
0.5	0.15	0.09	-0.01	-0.07	0.5	0.74	0.54	0.29	0.17
2.0		0.28	0.17	0.11	2.0		1.99	1.77	1.62
8.0			0.22	0.18	8.0			4.04	4.31
20.0				0.18	20.0				7.05
$\beta\hbar\omega_{\ddagger} = 4$					$\beta\hbar\omega_{\ddagger} = 12$				
$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0	$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0
0.5	0.29	0.18	0.03	-0.05	0.5	0.88	0.66	0.40	0.27
2.0		0.65	0.48	0.39	2.0		2.49	2.27	2.11
8.0			0.71	0.69	8.0			5.73	6.16
20.0				0.74	20.0				11.2
$\beta\hbar\omega_{\ddagger} = 6$					$\beta\hbar\omega_{\ddagger} = 16$				
$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0	$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0
0.5	0.44	0.29	0.10	0.00	0.5	1.18	0.93	0.63	0.49
2.0		1.06	0.86	0.74	2.0		3.53	3.33	3.15
8.0			1.51	1.53	8.0			9.65	10.4
20.0				1.81	20.0				21.2
$\beta\hbar\omega_{\ddagger} = 8$					$\beta\hbar\omega_{\ddagger} = 18$				
$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0	$\alpha_1 \backslash \alpha_2$	0.5	2.0	8.0	20.0
0.5	0.59	0.41	0.19	0.08	0.5	1.32	1.06	0.75	0.61
2.0		1.52	1.29	1.16	2.0		4.06	3.88	3.69
8.0			2.62	2.74	8.0			11.8	12.6
20.0				3.83	20.0				26.6

The results of Johnston and Heicklen inspired several other approaches that interpolate the energy along the reaction barrier based on properties of the reactants, products, and saddle point [24,30]. The table shows how extremely important tunneling corrections can become. Far beyond two- or ten-fold enhancements, tunneling at low temperatures can be responsible for thousand fold enhancements or more. Even at room temperature, large tunneling corrections are common. The importance of tunneling for room temperature enzyme reactions has been widely noted [31]. Garcia-Viloca et al. write: “In the transmission coefficient, recrossing events and nonequilibrium excitations appear to be unimportant, but tunneling can lead to a sizeable contribution to rate enhancements” [32,33]. For aromatic amine dehydrogenase, Masgrau et al. estimated that tunneling accounts for as much as 99.9% of room temperature reactivity [34].

12.2 Kinetic isotope effects

The potential energy surface in Cartesian coordinates is isotope independent, but isotopes do change the mass weighted landscape and the vibrational frequencies. A kinetic isotope effect (KIE) is a ratio between reaction rates with and without isotopic substitution. KIEs are particularly strong for protons (H) and deuterons (D) where the mass difference is a factor of two. For other elements, e.g. ^{12}C vs. ^{13}C , the isotopic mass differences are relatively small and accordingly their KIEs are more subtle. Figure 12.2.1 [35] illustrates how primary KIEs emerge from differences in ground state energies and tunneling probabilities [36].

Figure 12.2.1: Different isotopes give different zero point energies, different transition state frequencies, and different propensities for tunneling. All of these factors can influence the activation energy on an Arrhenius plot. [From Kohen, Prog. Reaction Kinetics and Mechanism, 28, 119–156 (2003).]



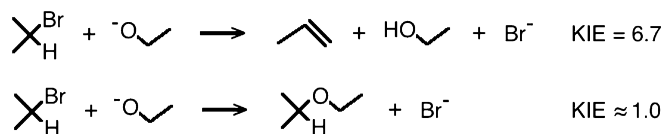
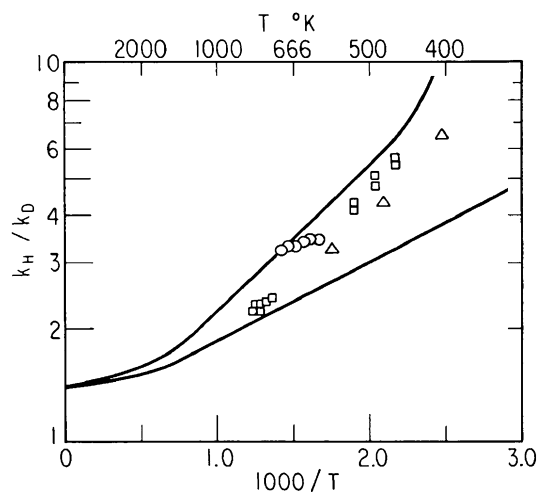
KIEs can be directly and somewhat reliably predicted from *ab initio* calculations. Errors in the computed potential energy surface mostly cancel from the KIE, leaving only ratios of more reliably predicted tunneling and zero point energy differences. In Johnston's words "The kinetic isotope effect is independent of small errors in calculated activation energy, and it is a sharp test of the vibrational analysis and tunneling correction" [25]. Figure 12.2.2 compares KIE data to two theoretical models [16], one including zero point effects and one including both zero point effects and tunneling effects.

KIEs are easily computed from the ratio of tunneling corrected TST rate expressions. Many of the partition functions in the ratio nearly cancel. Historically, factors in the KIE with values near unity were neglected entirely. These approximate calculations are useful for developing a chemical intuition for interpreting the magnitude of a KIE. An example calculation, adapted and somewhat extended from the text by Espenson [37], is shown below.

■ Example: KIEs from selectivity

The H vs. D kinetic isotope effect has been studied for each of the two reactions shown below [37]. The H atom adjacent to the initial $C - Br$ bond in the first reactant has been exchanged with deuterium.

Figure 12.2.2: KIEs for H or D atom abstraction from the CH_3 groups of acetone and ethane by a methyl radical. The lower curve includes differences in zero point energies, but not tunneling. The upper curve includes an Eckart barrier tunneling correction. Johnston noted that these results must be the effect of high temperature tunneling because the $C-H$ and $C-D$ vibrational frequencies are precisely known. [From Johnston and Rapp, *J. Am. Chem. Soc.* 83, 1–9 (1961).]



The absence of an appreciable KIE for the second reaction is not surprising, because the CH bond has not been broken. Because the second reaction has a negligible KIE, the KIE for the first reaction is evident from the product selectivity. No kinetics experiments are needed to estimate its KIE!

Espenson attributes the large KIE for the first reaction to a zero point difference between ground states because “the two transition states (where $R-H$ and $R-D$ are breaking) are at the same energy.” The transition states are indeed at the same energy, but tunneling may alter the activation energies. Explicitly accounting for tunneling and zero point energy factors in the KIE gives

$$\frac{k_H}{k_D} = \exp \left[\frac{\hbar(\omega_{CH} - \omega_{CD})}{2k_B T} \right] \cdot \frac{\kappa_{QM}(H)}{\kappa_{QM}(D)} \cdot (\text{secondary KIE}) \quad (12.2.1)$$

where all other factors have been lumped into a secondary KIE. Secondary KIEs arise from subtle frequency shifts in the unaltered bonds and from changes in rotational and translational partition functions. The secondary KIE can be obtained from a complete analysis of translation, rotation, and vibration with and without isotopic substitution. Usually the secondary KIE ratio is near unity.

The exponential term (part of the primary KIE) involves the difference in zero point energies of the bonds that are being broken. A back-of-the-envelope estimate of the zero point differences can be obtained from the harmonic oscillator frequency formula:

$$\omega = \sqrt{k/\mu}$$

where k is a force constant and μ is a reduced mass. The force constant is just a second derivative of the potential energy, so $k_{CH} = k_{CD}$. If we *approximate* that simple rules for effective masses apply within a big molecule, $\mu_{CH} = m_H m_C / (m_H + m_C)$, and likewise for μ_{CD} . C is already much heavier than H or D , so $\mu_{CH} \approx m_H$ and $\mu_{CD} \approx m_D$. Therefore,

$$\frac{\omega_{CH}}{\omega_{CD}} \approx \sqrt{\frac{m_D}{m_H}} = \sqrt{2}$$

The aliphatic CH bond has a typical stretching frequency of 2900cm^{-1} , which gives $\hbar\omega_{CH} = 34.67\text{kJ/mol}$. Dividing by $\sqrt{2}$ gives the corresponding CD stretch quanta $\hbar\omega_{CD} \approx 24.52\text{kJ/mol}$. Plugging these values into equation (12.2.1) for a temperature of 298K gives

$$\frac{k_H}{k_D} \approx 7.8 \cdot \frac{\kappa_{QM}(H)}{\kappa_{QM}(D)} \cdot (\text{secondary KIE})$$

7.8 is not far from the experimental value of 6.7, but our calculation was rather primitive. How should the result be interpreted? It is possible that the tunneling and secondary KIE factors are both nearly unity, that they partially cancel, or that our vibrational analysis in the primary KIE calculation is too simplistic. Without a full scale computational analysis of the rates with and without isotopic substitution, the KIE results should be interpreted with some caution. On a more optimistic note, the simple calculation has correctly identified a substantial KIE for one reaction and not for the other [37].

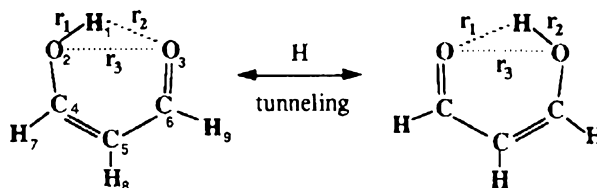
With modern *ab initio* calculations, it is easy to perform the full rate calculation including isotope effects on all frequencies, rotational partition functions, translational partition functions, and tunneling factors. By doing the calculation this way, one can be sure to account for both primary KIEs (those arising from isotope effects on the reaction coordinate degree of freedom), and secondary KIEs (those arising from effects on other degrees of freedom). Because of the relative ease with which KIEs can be computed, they have become an important link between experiment and theory. Our discussion has focused on KIEs for elementary reactions. However, they are perhaps most useful in understanding multistep mechanisms. When the reactants are labeled, the KIE can tell us whether the labeled atom is involved in the rate determining step. Moreover, experiments at varied reactant concentrations can have different rate determining

steps. KIEs in this situation can reveal which elementary step is overtaking the role of kinetic bottleneck as concentrations change [38].

12.3 Tunneling or tunnel splitting

Each of the tunneling corrections above assume a continuum of energy levels. How should tunneling be interpreted when the potential resembles a double-well with two bound states? Before Gamow's model [6] of alpha decay, Hund developed a theory for tunneling between bound states of molecules [39,40]. To this day, many studies of tunneling examine molecules that have symmetric double-well type potentials: tropolone, malonaldehyde, ammonia inversion, etc. [41–43]. Figure 12.3.1 depicts the two symmetric bound states of malonaldehyde [42].

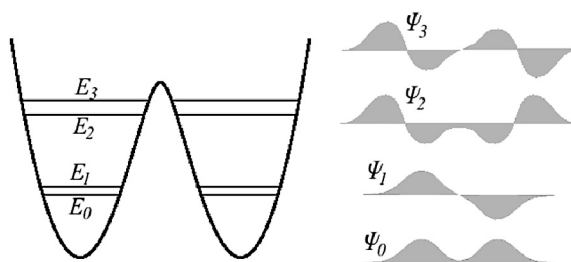
Figure 12.3.1: Intermolecular proton transfer in malonaldehyde involves tunneling through the barrier between quantized energy levels. [From Shida et al. *J. Chem. Phys.* 91, 4061 (1989).]



For a true one-dimensional symmetric double-well potential, the energy levels occur in narrowly separated pairs. Each pair corresponds to a state with amplitude on both the left and right hand side of the barrier as shown in Figure 12.3.2. The symmetric double-well requires that we abandon the model with a continuous stream of reflected and transmitted particles. Instead we must consider time-dependent quantum mechanics and state-to-state transitions. Suppose that at time $t = 0$, all amplitude is isolated on the left side of the double-well. All eigenstates have equal amplitude on the left and on the right, so the initial wavefunction is a mixture of energy eigenstates. If the initial condition is a mixture of just two narrowly separated states, then amplitude will shift from the left well to the right well and back with a period [4]

$$\tau = h / \Delta E \quad (12.3.1)$$

Figure 12.3.2: Symmetric double wells give discrete paired states with energies that differ by a narrow tunnel splitting energy ΔE . Symmetry allows the system to hop between the two wells faster than the rate predicted by tunneling estimates.



where ΔE is the tunnel splitting energy. The tunnel splitting energy, depicted in [Figure 12.3.2](#), can be estimated using the WKB approximation as

$$\Delta E = \frac{\hbar\omega_0}{\pi} \exp[-\theta(E)] \quad (12.3.2)$$

Here ω_0 is the frequency in the stable wells and $\theta(E)$ is the same barrier penetration integral defined in [section 12.1](#). The value of E is intermediate between the two split energy levels. An energy that straddles the split pair of energy levels can also be identified by the WKB approximation. The (approximate) energy of the n^{th} pair is given by the root E_n for the equation

$$\frac{1}{\hbar} \int_{q_1(E_n)}^{q_2(E_n)} dq \sqrt{2m(E_n - V(q))} = (n + 1/2)\pi \quad (12.3.3)$$

where the action integral spans the classically *allowed* region in one well, i.e. where $q_1(E_n)$ and $q_2(E_n)$ are the left and right classical turning points at energy E_n . For a one-dimensional model, the effects of other states can be ignored if their energy levels are far from those of the narrowly split pair [\[44\]](#). Tunnel splittings can be probed experimentally, e.g. by microwave and NMR spectroscopy, so “double-well” tunneling processes provide unusually direct tests for computational predictions. For malonaldehyde, one dimensional *ab initio* models provide reasonable approximations to the observed tunnel splitting [\[41\]](#).

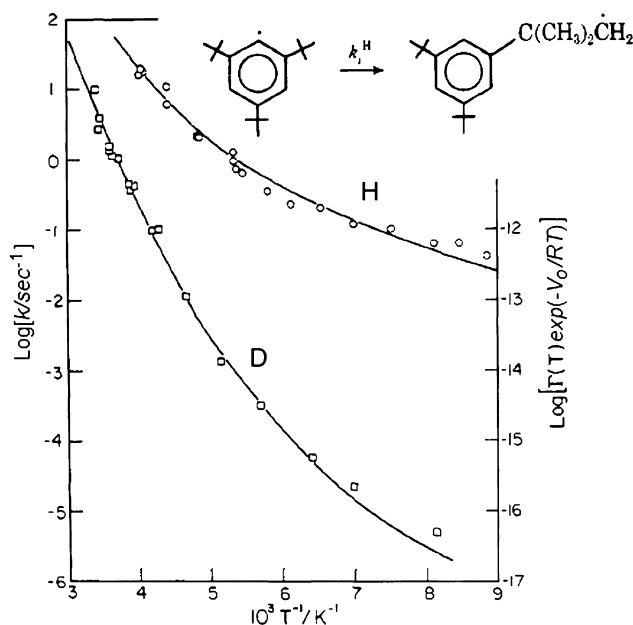
Tunnel splittings can also be computed from multidimensional models, although the calculations are significantly more difficult [\[43,45\]](#). Fortunately, multidimensional molecular potential energy surfaces tend to have closely spaced and vibrationally coupled energy levels. These can often be treated as a continuum of energy levels, which allows the use of simple tunneling corrections like those in [section 12.1](#). [Figure 12.3.3](#) shows an example in which the Eckart tunneling correction accurately describes hydrogen transfer between two bound sites on a single molecule [\[46\]](#).

Successful results like that shown in [Figure 12.3.3](#) must still be interpreted carefully. Computational KIE estimates are somewhat reliable, but absolute rate calculations are not. In cases of agreement, it is difficult to tell whether all parts of the absolute rate calculation are working correctly, or whether some errors are fortuitously compensating for others.

12.4 Multidimensional tunneling models

Recall that one dimensional tunneling models assumed dynamical separability of the reaction coordinate. Curved reaction pathways couple the reaction coordinate to transverse vibrational modes [\[47\]](#), causing assumptions of the one-dimensional tunneling models to break down. Note that curvature does not require atoms to follow a physically curved trajectory in space.

Figure 12.3.3: Intermolecular hydrogen transfer between aryl radical and alkyl radical states must involve a double well type potential, but is adequately described by an Eckart tunneling correction based on continuous energy levels. Note the extremely large kinetic isotope effect. [From Brunton et al. J. Am. Chem. Soc. 98, 6803-11 (1976).]



Even the collinear atom transfer reactions have curved reaction pathways because the changes in bondlengths are not perfectly concerted. Hydrogen transfer reactions with curved reaction pathways exhibit corner cutting tunnel paths and other intrinsically multidimensional tunneling mechanisms. This section provides a very brief (and vastly oversimplified) view of multidimensional tunneling theories.

Small curvature tunneling models

Small curvature tunneling models account for curvature by computing the barrier penetration integral along approximately optimal corner-cutting tunnel paths [19,48,49]. The tunneling probability and the barrier penetration integral are computed using the one dimensional formulas, $P_{SC} = (1 + \exp[2\theta(s)])^{-1}$, and

$$\theta(s) = \frac{1}{\hbar} \int_{s_1}^{s_2} \sqrt{2\mu(s) |V(\mathbf{x}(s)) - E|} ds.$$

However, the action is computed for a multidimensional curve $\mathbf{x}(s)$ that does not follow the MEP. In these calculations, the path that minimizes the action maximizes the tunneling probability. Based on the structure of the action integral, the optimal path will be determined by a compromise between arclength and energy along the path. Paths that cut corners are short, but paths that follow the MEP traverse the lowest energy routes over the barrier.

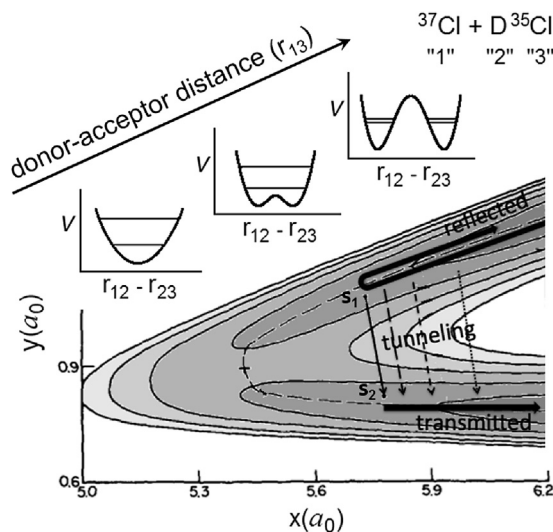
Marcus and Coltrin [48] considered curvature effects in tunneling corrections for $\mathbf{A} + \mathbf{BC} \rightarrow \mathbf{AB} + \mathbf{C}$ reactions. They treated the transverse mode as an adiabatic ground state vibration at each point along the reaction pathway. They noted that the inner envelope of the vibrations naturally reduces the tunneling path length without increasing the energy level [19,48]. Truhlar, Garrett, and coworkers generalized the Marcus-Coltrin treatment to more complicated reactions in their small curvature tunneling theory [19,49]. In addition to properties required for a harmonic TST calculation, small curvature tunneling calculations require the vibrationally adiabatic ground state potential energy profile along the MEP and hence transverse vibrational frequencies along the MEP. Small curvature tunneling has provided accurate results for several cases [34,50], but there are also some indications that it overestimates the effects of curvature [26,51].

Large curvature tunneling models

Transfer of the hydrogen or proton between heavy donor and acceptor centers, e.g. in the $\text{Cl} + \text{HCl} \rightleftharpoons \text{Cl} + \text{HCl}$ reaction, leads to a number of surprising deviations from classical TST. Recall that heavy-light-heavy atom transfer reactions have small skew angles and sharply curved transition pathways. Babamov and Marcus noted that the donor-acceptor approach motion is slow while the H -atom transfer between the donor and acceptor atoms is fast. They introduced coordinates to separate these motions and to build a non-adiabatic model. Their analysis showed that zero point energy in the H -transfer degree of freedom causes the entrance and exit channels to merge before the donor and acceptor distances reach the saddle point [52]. Their results revealed a strong “corner cutting” behavior in which the H -atom jumps from one channel to the other well before reaching the saddle point location.

Garrett et al. developed a large curvature tunneling model [53,54] which can be applied to more complex heavy-light-heavy reactions. Garrett and Truhlar assume that trajectories approach the saddle region adiabatically from $s = -\infty$ up to an arclength $s_1 < 0$ along the MEP. The arclength s_1 is that for which the vibrationally adiabatic ground state energy equals the incident energy, i.e. s_1 is determined by $E = V_0(s_1) + \sum_{\text{transverse}} \hbar\omega_i(s_1)/2$. The equation for s_1 has a second root that defines the other turning point $s_2 > 0$ [54]. A trajectory coming from $s = -\infty$ has opportunities to tunnel across the ridge to the other channel at each value of s leading up to s_1 . If no tunneling occurs, it has a second opportunity to tunnel across the ridge as the donor and acceptor fragments recede back to $s = -\infty$. Trajectories with sufficiently high incident energy E can also jump from one channel to another without tunneling. Several variations of the large curvature tunneling method have been developed [30,55,56], with each version approximating the tunnel path from s_1 to s_2 in a different way. Figure 12.4.1 illustrates the original method of Garrett et al. in which a straight-line tunneling path was adopted. Figure 12.4.1 also shows (in the insets above) the non-classical corner cutting mechanism of Babamov and Marcus.

Figure 12.4.1: The large curvature tunneling correction approximates the tunneling probabilities as the donor and acceptor atoms approach (and retreat) at energy E in a heavy-light-heavy reaction. The insets above show how donor-acceptor distance modulates the potential for the H-transfer step, and in particular how zero point energy facilitates the H-transfer even for trajectories that do not reach the saddle point region. [Modified from Garrett et al., *J. Chem. Phys.* 78, 4400-13 (1983).]



The semiclassical transition state theory (SCTST) framework by Hernandez and Miller used spectroscopic constants to account for anharmonicity, reaction pathway curvature, and tunneling in the vicinity of the saddle point. Additionally, the SCTST formalism accounts for tunneling via all possible paths. Some applications suggest these methods can give accurate low temperature results [26,57], but the anharmonicity inputs are not easily obtained.

Exercises

1. Section 12.1 notes that the truncated parabolic barrier model correctly predicts the vanishing of activation energy at low temperature. Show how this happens by constructing the entire rate constant $k = k_{TST}\kappa_{QM}$ with κ_{QM} from Equation (12.1.22).
2. Reproduce the square top barrier results from Figure 12.1.3. Then create similar plots using the WKB approximation (Equations (12.1.14)–(12.1.16)). How do the exact results compare to those from the WKB approximation?
3. Equilibrium isotope effects are changes in the equilibrium constant due to isotopic substitution. Explain how KIEs for the forward and reverse reactions are related to the equilibrium isotope effect.
4. Develop a one-dimensional tunneling correction for the infinite triangular barrier: $V(x) = V_{\ddagger} - A|x|$. Where might this correction be applicable?
5. Perform an *ab initio* calculation to obtain the geometries, energies, and frequencies of the reactant and transition state configurations for the $H_2 + H \rightarrow H + H_2$ reaction. Compute the TST rate constant and the rate with an Eckart barrier tunneling correction. Construct

Eyring plots from the TST rate and from the tunneling corrected TST rate at three temperatures. How much does tunneling affect the apparent activation enthalpy and entropy?

6. RRKM theory can be extended to include the tunneling probability $P_{QM}(E)$ at each energy E . Let E_0 be the threshold energy, $\rho(E)$ the density of states at energy E , and $N^\ddagger(E - E_0)$ the number of ways to partition energy beyond the threshold among the non-reaction coordinate degrees of freedom at the transition state. The tunneling corrected RRKM formula is

$$k_{RRKM-QM}(E) = \frac{N_{QM}^\ddagger(E - E_0)}{h\rho(E)}$$

Here,

$$N_{QM}^\ddagger(E) = \int_0^E P_{QM}(E_t) \rho(E - E_t) dE_t$$

is a tunneling probability weighted cumulative density of states up to energy E .

- Replace $P_{QM}(E)$ with the Heaviside function $\theta(E - E_0)$, corresponding to perfect transmission (unity) at energies above the threshold and zero transmission at energies below. Show that the RRKM theory is recovered.
- Sketch $N_{QM}^\ddagger(E)$ and its ‘classical’ counterpart $N^\ddagger(E)$ vs. E/E_0 . See Gray et al. for an application to the unimolecular reaction $HNC \rightarrow HCN$ [58].

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